

Puszka Pandory

MIW Final Report — s31880

1. Reverse engineering

1.1. The original game

Puszka Pandory was written in 1986 by Marcin Borkowski for the ZX Spectrum 48K. Borkowski coded it in assembler. The game started as a text compression experiment and turned into a full adventure along the way. It was inspired by Marek Baraniecki's novel "Glowa Kasandry."

The map is 40 by 40 locations, which is large for 48 kilobytes of RAM. The player types commands in Polish — roughly 20 words the parser recognizes. A small graphic showing the current terrain sits in the center of the screen. There is no save function. If you die, you restart from the beginning.

According to the Polygamia article by Bartłomiej Kluska, the game had no graphics and no music beyond a brief funeral march. Kluska spent roughly 15 hours mapping it by hand. He described the vocabulary as weak, the descriptions as dry, and the plot as thin compared to Western adventures from Infocom and similar studios.

1.2. Publication

Borkowski first tried selling the game at the Grzybowska computer market in Warsaw. It did not sell. He later transferred distribution rights to ENTER Computing, which made it the first legally sold full-scale adventure game for ZX Spectrum in Poland.

In April 1989, Bajtek magazine (issue 40, pages 16-17) ran a two-page review. It was the first time any Polish computer game got that kind of coverage in the magazine. The reviewer was listed as "Karol B. Mirowski" — actually Borkowski himself under a pseudonym. The review noted the large map but criticized the sparse graphics, lack of sound, and missing save function.

Bartłomiej Kluska later wrote that Puszka Pandory was "the first computer game by a Pole that was attempted to be sold in Poland and received a review in a Polish publication."

A collector's edition was released in 2012 at Poznan Game Arena, packaged on a cassette in a metal case. An original 1986 copy went for 3,306 zloty at a WOSP charity auction in 2018.

1.3. Story

Post-apocalyptic setting. A third world war has happened. On the coast of South America, survivors have rebuilt a city. A missile system called Pandora's Box, built by a minor nation, threatens to destroy what is left. The player is a special agent dropped onto an unnamed island in the Pacific to disarm it. The helicopter gets shot down on approach. The player lands by parachute.

1.4. What I found during play-testing

I ran the game in Speccy on Windows. You start in a meadow. Movement is compass directions. The map splits into two main routes: north through forest and jungle toward a volcano, or east to the coast and a cave system.

The puzzle chain:

- Gas mask (swamp) is needed to survive the volcano fumes.
- A key (waterfall) or crowbar (cove) opens the bunker door.
- Planks (ruins) fix the bridge to the missile silo.
- Wire cutters (hill) or explosives (cave) disable the missile.
- Signal flare (volcano slope) calls extraction from the crater rim after the missile is down.

Some locations kill you for going the wrong direction: the crater rim, open sea, waterfall ledge, missile silo grating. The bridge kills you if you step south or east off it. No warning. These directions are not even listed as exits.

Other items: shovel in the meadow (dig twice for a buried tin box), rations on the beach, radio in the rocky field, journal in the ruins with handwritten hints.

The backpack holds four items, not more. At several points in the critical path I needed more than four things simultaneously, which meant dropping items, walking back to retrieve them, and replanning routes. Deaths here were the most frustrating because you restart with everything scattered.

Forest is the dominant terrain on the map. Most of it is fine, but any tile surrounded by forest on all four sides triggers random movement: each step goes in a direction the game picks, not you. The only way out is to use the compass, which sits near D-3 by the lake. Without it the forest is not navigable. I spent close to three hours mapping safe corridors before I understood this mechanic was intentional.

Two more hazards in the middle section: a minefield near the lake and a magnetic anomaly. The mines are placed randomly each run, which means knowing the safe path from one attempt does not help on the next. The mine detector beeps when a mine is one tile away and needs to be switched on manually before it works. The magnetic anomaly disables the compass, which drops you back into random movement until you leave the field.

The second half of the game requires an aqualung and an oxygen apparatus to enter an underwater complex. The controls change there: O is right, I is forward, U is left. There is an oxygen countdown. The layout is large enough that I had to draw it on paper. I died twice before getting through.

The ending is not signposted inside the game. At the control room you type MÓWIC (speak) followed by a password. PANDORA triggers the bad ending. KASANDRA is correct — the name comes from the Baraniecki novel the game is based on. The only published hint pointed to a 1983 issue of *Fantastyka* magazine.

1.5. Sources

- [Puszka Pandory — Wikipedia \(pl\)](#)
- B. Kluska, “Opowiesci z krypty: Puszka Pandory” — *Polygamia*
- [Odyseja Przygody — collector’s edition photos and cassette artwork \(2012 reissue\)](#)
- [Gra.pl — “Historia polskich gier: poczatki i Puszka Pandory”](#)
- [SWI-Prolog download](#)

2. Recreation

2.1. How to run

Written in SWI-Prolog 10. If SWI-Prolog is not installed:

- Windows: download from swi-prolog.org/download/stable and run the installer.
- Linux (Debian/Ubuntu): `sudo apt install swi-prolog`
- Linux (Arch): `sudo pacman -S swi-prolog`
- macOS: `brew install swi-prolog`

Then:

```
swipl -s s31880MIWFINAL.pl
?- start.
```

No external libraries. Type `help.` in-game for commands.

2.2. How it works

Game state lives in Prolog's dynamic database: `player_at/1`, `item_at/2`, `holding/1`, `world_state/2`, `alive/0`, `game_won/0`. The predicate `init/0` wipes everything and rebuilds from `initial_item/2` facts. Location descriptions are predicates (`desc/1`), not stored strings — `show_state/1` adds conditional text depending on what the player has changed.

The game loop reads a term with `read/1`, dispatches it through `do/1`, and recurses until `alive` is retracted or `game_won` is set.

The original has 1,600 locations. This version has 23. They cover the main areas: island interior, coast, cave and bunker system, missile silo.

All 23 locations have a `location_pos(Name, Row, Col)` fact that places them on a 7-row by 8-column grid. This makes the map a proper 2D structure: adding a new location requires one `connected/3` fact (for movement) and one `location_pos/3` fact (for the grid). No other predicates need changing.

2.3. Minimum requirements

2.3.1. 1. Movement

Commands: `north`, `south`, `east`, `west` (or `n`, `s`, `e`, `w`). 23 locations connected by `connected/3` facts. Three transitions are gated:

```
try_move(cave_interior, n, bunker_door) :- !,
    (world_state(bunker_door, unlocked) -> do_move(bunker_door)
    ; write('Locked. Find a key or something to break it.'), nl).
```

```
try_move(old_bridge, n, missile_silo) :- !,
    (world_state(old_bridge, reinforced) -> do_move(missile_silo)
    ; write('Too many gaps. Fix the planks first.'), nl).
```

```
try_move(volcano_slope, n, crater_rim) :- !,
    (holding(gas_mask) -> do_move(crater_rim)
    ; write('Sulfur fumes. You choke. Need a mask.'), nl).
```

The original game also gates these transitions the same way: the volcano requires a mask, the bunker requires a key, the bridge requires repair.

2.3.2. 2. Looking around

`look` (or `l`) prints the location name, description, world state changes, items on the ground, exits, and then the ASCII island map with the current position marked. `show_state/1` handles conditional text: the bunker shows a countdown clock when the missile is armed, and “Console wrecked” after sabotage. The cave interior reports whether the steel door is locked or open.

In the original, looking around also listed terrain, items, and exits, and displayed a small terrain graphic in the center of the screen. The map display in this version serves the same role.

2.3.3. 3. Inventory

`take(Item)` moves an item from `item_at/2` to `holding/1`. `drop(Item)` does the reverse. `inventory` (or `i`) lists carried items. Missing items get a specific error message.

The original used Polish commands (“WZIAC” for take). This version uses English.

2.3.4. 4. Permanent world changes

Digging in the meadow asserts `world_state(meadow, dug)`. Digging again changes it to `world_state(meadow, deep_hole)` and spawns a tin box. If the player leaves and comes back, `show_state/1` still prints the hole description. Same pattern for unlocking the bunker door, fixing the bridge, and disabling the missile.

```
dig_at(meadow) :- !,
    dig_count(N), retract(dig_count(N)),
    N1 is N + 1, assert(dig_count(N1)),
    (world_state(meadow, dug) ->
        retract(world_state(meadow, dug)),
        assert(world_state(meadow, deep_hole)),
        write('Deeper. Something metal down there. A tin box.'), nl,
        assert(item_at(tin_box, meadow))
    ); assert(world_state(meadow, dug)),
    write('Hole dug. Nothing yet.'), nl.
```

Digging the same spot a third time caves in the walls and kills the player. In the original, items also stayed where you dropped them and world changes persisted.

2.3.5. 5. Biomes

23 locations: meadow, forest edge, deep forest, jungle, swamp, volcano slope, crater rim, riverbank, stream, waterfall, beach, sea shallows, cove, cave entrance, cave interior, bunker, sandy trail, hill, rocky field, cliff base, ruins, old bridge, missile silo. This covers every terrain type found during play-testing of the original.

2.4. Beyond minimum requirements

2.4.1. Death

Six ways to die. Dig the meadow three times: cave-in. Dig the volcano slope: steam crack. Walk the wrong direction off the crater rim, sea shallows, waterfall ledge, or silo grating: instant death. Detonate C4 in the wrong place: explosion. The original also killed the player for wrong moves in dangerous areas — Kluska mentions losing hours of progress to unexpected deaths with no save function.

```
blocked_move(crater_rim, Dir) :- Dir \= s, !,
    write('You go over the rim. Magma. That is it.'), nl, die.
```

2.4.2. Two endings

Firing the signal flare from the crater rim ends the game. If the missile has been sabotaged, `win/0` sets `game_won` and removes `alive` without printing GAME OVER. The player is extracted by helicopter. If the missile is still armed, the helicopter comes anyway, but Neo City is destroyed three days later. The game prints MISSION FAILED and calls `die`.

The original also has two endings, but they work differently: the player types MÓWIĆ KASANDRA or MÓWIĆ PANDORA at the control room. Here the two endings come from whether the missile is disabled before firing the flare, not from a password input.

2.4.3. Storyline

The game opens with a backstory. The Cuban Missile Crisis did not end — the bombs fell in October 1962, billions died, and the survivors rebuilt on the South American coast. A Soviet auto-launch system on a remote island is still running. The player is a CIA agent sent to stop it.

The original uses a post-WWIII setting drawn from Baraniecki’s novel. I swapped that for a 1962 nuclear war because it gave a sharper historical anchor without requiring the player to know the source material. The bones are identical: agent, island, missile, extraction.

Three short narrative messages appear the first time the player enters the cave interior, the bunker, and the silo. The original has nothing like this.

2.4.4. Readable items

`read(journal)` and `read(map_fragment)` print in-game hints when the player holds the item. If the player does not hold it, the game says so:

```
do(read(journal)) :- holding(journal), !,  
    nl, write('"Day 14. Missile is real. Silo is north of the bridge. '), nl,  
    write(' Bunker console runs the launch. Cut the cables or wreck it. '), nl,  
    write(' Fumes stopped me at the volcano. Mask must be in the swamp. '), nl.  
do(read(journal)) :- !, write('You do not have the journal. '), nl.
```

The journal is in the ruins, reachable before the player needs the gas mask. The hints arrive before the problems they describe.

2.4.5. Alternative puzzle solutions

The bunker door opens with the key (waterfall) or the crowbar (cove). The missile is disabled with wire cutters (hill) or C4 (cave interior). Two routes through the core puzzle, two tools for each gate.

2.4.6. Blocked direction messages

About 20 impossible directions have individual messages instead of a generic “You cannot go that way.” East from the riverbank: “River is wide and fast here. No crossing.” South from the swamp: “Deeper water south. Not going there.” North off the crater rim: death. The original also returned specific messages for out-of-bounds moves.

2.4.7. Hidden item: `tin_box`

Digging the meadow twice uncovers a buried tin box. `use(tin_box)` opens it and reveals a note with hints about the bunker, the bridge, and the gas mask. This is a second hint source for players who explore the meadow before finding the journal in the ruins.

2.4.8. ASCII map and 2D grid

Every `look` and `map.` command renders the island as a 7-row by 8-column ASCII grid. Each cell shows a two-letter location code. The current position is marked with `*`. Empty cells show `.`. The layout below shows the map as it appears at game start, with the player in the meadow (`*md`):

```
MAP   0   1   2   3   4   5   6   7
      +---+---+---+---+---+---+---+
0     | . | . | cr| . | . | . | ms| . |
      +---+---+---+---+---+---+---+
1     | . | . | vs| . | . | ru| br| . |
      +---+---+---+---+---+---+---+
2     | . | . | jn| sw| . | . | . | . |
      +---+---+---+---+---+---+---+
3     | . | . | df| . | st| wf| . | . |
      +---+---+---+---+---+---+---+
4     | . | cb| fe| . | rb| bc| cv| ce|
      +---+---+---+---+---+---+---+
5     | . | rf|*md| tr| . | ss| . | ci|
      +---+---+---+---+---+---+---+
6     | . | hl| . | . | . | . | . | bn|
      +---+---+---+---+---+---+---+
```

`*` = your position

Location codes: `cr` crater rim, `ms` missile silo, `vs` volcano slope, `ru` ruins, `br` bridge, `jn` jungle, `sw` swamp, `df` deep forest, `st` stream, `wf` waterfall, `cb` cliff base, `fe` forest edge, `rb` riverbank, `bc` beach, `cv` cove, `ce` cave entrance, `rf` rocky field, `md` meadow, `tr` sandy trail, `ss` sea shallows, `ci` cave interior, `hl` hill, `bn` bunker.

The original game displayed a small horizontal terrain graphic in the center of the screen that updated with each move. The ASCII map here plays the same role. The `map.` command shows it without re-printing the room description. `use(compass).` also triggers the map display.

The 2D grid is stored as `location_pos/3` facts. The renderer iterates `between(0,6,Row)` and `between(0,7,Col)`, calls `print_map_cell/3` for each coordinate, and looks up the location by position:

```
print_map_cell(Row, Col, Here) :-
    ( location_pos(Loc, Row, Col) ->
        map_abbrev(Loc, Ab),
        ( Loc == Here -> format("*~w|", [Ab]) ; format(" ~w|", [Ab]) )
    ; write(' . |') ).
```

2.4.9. Additional items: compass and rope

The original had a compass (coordinates D-3 near the lake) and a rope. Both appear in walkthrough sources. I included both but changed what they do.

The compass sits in the rocky field next to the radio. In the original it prevented forest disorientation; since our version has no random movement, `use(compass).` shows the ASCII map from wherever you are.

The rope is at the waterfall. The location description says “something dark behind the curtain,” which is where it turns up. It reinforces the old bridge the same way planks do:

```
use_item(rope, old_bridge) :- \+ world_state(old_bridge, reinforced), !,  
    write('You lash the rope across the gaps as a lifeline. It will hold. '), nl,  
    retract(holding(rope)), assert(world_state(old_bridge, reinforced)).
```

The bridge now has two solutions to match the bunker door (key or crowbar) and the missile (wire cutters or C4).

2.5. Changes from the original

2.5.1. Commands and language

The original parser understood roughly 20 Polish words, all requiring correct diacritics. WZIAĆ was take. ŚCIAĆ was cut. ZAŁOŻYĆ was put on or wear. PLYNAĆ was swim, typed before a direction when in water. MÓWIĆ was speak, required for the final puzzle. This version uses English Prolog terms: `take(item)`, `use(item)`, directional atoms for movement. There is no speak command.

2.5.2. Inventory limit

The original capped the backpack at four items. This version has no limit.

2.5.3. Water movement

In the original, moving through water required typing PLYNAĆ before the direction. Here directional commands work the same on land and in water.

2.5.4. Forest disorientation

The original's main navigation hazard: stepping onto a forest tile surrounded by forest on all four sides caused random movement until the compass was used. This version uses deterministic movement throughout, so the hazard is absent.

2.5.5. Compass

In the original the compass (found at D-3 near the lake) prevented forest disorientation and stopped working in the magnetic anomaly field. Here the compass shows the ASCII map. The item is present, its purpose is different.

2.5.6. Mines and mine detector

The original had a minefield near the lake with randomly placed mines each run. The mine detector gave audio feedback when a mine was adjacent and had to be switched on manually. Both are absent here because the minefield section was not implemented.

2.5.7. Magnetic anomaly field

A terrain type in the original that disabled the compass and caused random movement. Not implemented.

2.5.8. Underwater section

The second major area of the original required an aqualung and oxygen apparatus, used a separate control scheme (O right, I forward, U left), and ran an oxygen countdown. Not implemented.

2.5.9. Final puzzle

The original ended at the control room: type MÓWIĆ KASANDRA for the good ending, MÓWIĆ PANDORA for the bad one. KASANDRA comes from the Baraniecki novel, but the game does not tell you this — the only documented hint pointed to a 1983 issue of *Fantastyka*

magazine. This version ends differently: disable the missile by any means, reach the crater rim, fire the flare. No password, no MÓWIĆ command. The two endings here (mission complete vs. mission abandoned) are a different design, not a recreation of the original's.

2.5.10. Items not implemented

The original contained an aqualung, oxygen apparatus, mine detector, and axe. All four are absent because the sections that used them were not implemented. A screwdriver also appeared in the original; walkthrough sources say it served no function.

2.5.11. Items added

Tin box (dug from the meadow), journal, and map fragment are not in the original. They were added as in-game hint sources. The rope is present in the original but used here differently (bridge reinforcement rather than whatever the original intended).

2.5.12. Digging

The original's shovel worked across many terrain types in ways I could not fully reconstruct. This version limits `dig` to four locations: meadow, beach, rocky field, and volcano slope. Each has a specific outcome rather than a generic “nothing here” for most tiles.

2.5.13. Scoring

The original had a points system. This version uses binary win/loss.

2.5.14. Map scale

1,600 locations versus 23. The original's sparse map was deliberate — most tiles had no items or descriptions. A full recreation was not attempted.

2.5.15. Story setting

The original uses a post-WWIII world based on Baraniecki's novel “Głowa Kasandry.” This version uses a 1962 Cuban Missile Crisis setting. The structure is the same (agent, island, missile, extraction) but the historical premise is different.

2.5.16. Graphics and sound

The original displayed a scrolling terrain strip on screen and played a funeral march when you died. This version has neither. The ASCII grid map takes the functional role of the terrain strip.

2.5.17. Save function

Neither version has one. `init/0` resets everything from scratch.